

Smart IDs: A Look At The Whole Tech Jigsaw

Comprising microcontrollers, memory and the like, a smart card is just one aspect of a whole smart ID setup. The other pieces of the jigsaw include smart card terminals, application software, communication networks and more. Let us take a quick look at all the tech constituents of a smart ID setup and the trends concerning these

A smart card is essentially a card or token comprising a secure microprocessor core and non-volatile memory. Being capable of identification, authentication, communication, application processing and data storage, it is amenable to functions ranging from credit, cash and insurance cards to health, travel and even SIM cards. Shiv Turmari, country sales manager, ARM Embedded Technologies, points out that now smart cards are being deployed even for lottery metering and vending, pay TV, gaming, GSM, healthcare, transportation and pay phones.

If done right, every person would need to carry just one card! "A smart card can hold multiple application data securely without any interference between the service applications. Hence, multiple services can be offered through a single smart card," explains Dayakar C. Reddy, managing director, MosChip Semiconductor.

However, the smart card by itself plays a very small role in this scheme—smart card terminals, communication networks and application software too play a large role.

Smart card basics

A smart card comprises a secure microprocessor core, memory and some means of powering its functioning.

"Of late, all these components have undergone changes from the price, performance, area and price points of view," says Turmari. Processors are constantly evolving to meet the rapidly evolving applications, increasing memory and speed requirements, strict cost control, and reduced power consumption needs of the industry.

Security. Speaking of a smart card's components, the word 'secure' requires special attention. Security, be it in the form of PIN numbers, passwords or encryption keys, can be compromised by smart hacking. A secure smart card tries to overcome this challenge by roping together several technologies including even high-level ones like biometric identification.

Sumit Sahai, product manager—processor division, ARM, explains that normal processors offer little or no security in this regard. "With sophisticated hacking equipment, smart card data can be intercepted, modified and spied upon during standard CPU operations. A secure processor core, on the other hand, would be tamper-resistant and prevent retrieval or modification of on-chip information even during the most sophisticated physical attacks. It should also be designed to fail gracefully, where the compromise of one

Market requirements and manufacturers' answer to these

Market trends and requirements:

- Demand for high performance stretching beyond existing 8-/16-bit proprietary cores
- Legacy proprietary architectures nearing end of life
- Mid-range SIM memory sizes increasing beyond 16-bit addressability
- Service consolidation—multiple services/applications running on one device
- Advanced security algorithms
- High performance at very low power
- Preference for NFC and contactless cards
- Increasing pressure on profit margins

Solution for the current market trends:

- 32-bit power to handle multiple services and advanced security algorithms
- 32-bit memory addressing
- Industry standard architecture
- Configurable security features
- Fast multiply and divide
- Easy designing
- Broad tools and ecosystem support
- Fast time-to-market
- Low-cost project entry

—Shiv Turmari of ARM

device does not compromise the entire system. The security features should also be highly configurable," he says.

Memory requirements. More memory is needed to cater to both the increasing complexity of applications and the tight security measures. Security algorithms such as 256-bit+ AES require larger keys, biometric identification involves large data sets, and operating systems and file systems for secure chips are quite big. Sahai mentions that such memory requirements are met using higher code-density and 32-bit addressing in current generation of microprocessors.

Power. The functioning of a smart card, especially the communication task, requires power. While some cards require contact to source power, some contactless ones (certain models of near-field communication (NFC) smart cards and radio frequency identification (RFID) smart cards) are capable of drawing power from the terminal itself during the read operation. The mode of powering a smart card, however, depends largely on the functions performed and the